

Mathematical Physics

Complete Study Guide for GATE 2026

Foundation for Success

Comprehensive Coverage of All Topics

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1 Introduction

Important for GATE

Mathematical Physics is the foundation of all physics topics in GATE. Mastering this ensures success in Quantum Mechanics, Classical Mechanics, Electrodynamics, and Statistical Mechanics. This guide covers every essential topic with no stone left unturned.

1.1 How to Use This Guide

1. Study each section thoroughly with understanding, not memorization
2. Solve all examples provided
3. Practice problems after each major topic
4. Revise formulas regularly
5. Focus on physical interpretation
6. Time yourself on practice problems

2 Vector Calculus

2.1 Vector Fields and Scalar Fields

Definition 2.1. A scalar field assigns a scalar value to each point in space: $\phi(\vec{r}) = \phi(x, y, z)$

A vector field assigns a vector to each point in space: $\vec{F}(\vec{r}) = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$

2.2 The Gradient (grad)

Definition 2.2. The gradient of a scalar field ϕ is a vector field defined as:

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

Important for GATE

Physical Interpretation:

- $\nabla\phi$ points in the direction of maximum increase of ϕ
- $|\nabla\phi|$ gives the rate of maximum increase
- $\nabla\phi$ is perpendicular to level surfaces $\phi = \text{const}$

Formula 2.3 (Gradient in Different Coordinate Systems). **Cartesian:**

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

Cylindrical (r, θ, z) :

$$\nabla\phi = \frac{\partial\phi}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial\phi}{\partial\theta}\hat{\theta} + \frac{\partial\phi}{\partial z}\hat{z}$$

Spherical (r, θ, ϕ) :

$$\nabla\phi = \frac{\partial\phi}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial\phi}{\partial\theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial\phi}{\partial\phi}\hat{\phi}$$

Example 2.4. Find $\nabla\phi$ where $\phi = x^2y + xyz - z^3$.

Solution:

$$\nabla\phi = (2xy + yz)\hat{i} + (x^2 + xz)\hat{j} + (xy - 3z^2)\hat{k}$$

2.3 The Divergence (div)

Definition 2.5. The divergence of a vector field $\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$ is:

$$\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Important for GATE**Physical Interpretation:**

- $\nabla \cdot \vec{F} > 0$: Source (fluid flowing out)
- $\nabla \cdot \vec{F} < 0$: Sink (fluid flowing in)
- $\nabla \cdot \vec{F} = 0$: Solenoidal/Incompressible field

Formula 2.6 (Divergence in Different Coordinate Systems). **Cartesian:**

$$\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Cylindrical:

$$\nabla \cdot \vec{F} = \frac{1}{r} \frac{\partial(rF_r)}{\partial r} + \frac{1}{r} \frac{\partial F_\theta}{\partial \theta} + \frac{\partial F_z}{\partial z}$$

Spherical:

$$\nabla \cdot \vec{F} = \frac{1}{r^2} \frac{\partial(r^2 F_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sin \theta F_\theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi}$$

Example 2.7. Find $\nabla \cdot \vec{F}$ where $\vec{F} = x^2 \hat{i} + y^2 \hat{j} + z^2 \hat{k}$.

Solution:

$$\nabla \cdot \vec{F} = \frac{\partial(x^2)}{\partial x} + \frac{\partial(y^2)}{\partial y} + \frac{\partial(z^2)}{\partial z} = 2x + 2y + 2z$$

2.4 The Curl

Definition 2.8. The curl of a vector field $\vec{F}$ is:

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} \\ &= \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \hat{i} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \hat{j} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \hat{k} \end{aligned}$$

Important for GATE**Physical Interpretation:**

- Measures rotation/circulation of a vector field
- $\nabla \times \vec{F} = 0$: Irrotational/Conservative field
- $|\nabla \times \vec{F}|$: Magnitude of rotation
- Direction: Axis of rotation (right-hand rule)

Formula 2.9 (Curl in Different Coordinate Systems). **Cylindrical:**

$$\nabla \times \vec{F} = \left(\frac{1}{r} \frac{\partial F_z}{\partial \theta} - \frac{\partial F_\theta}{\partial z} \right) \hat{r} + \left(\frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r} \right) \hat{\theta} + \frac{1}{r} \left(\frac{\partial(rF_\theta)}{\partial r} - \frac{\partial F_r}{\partial \theta} \right) \hat{z}$$

Spherical:

$$\begin{aligned} \nabla \times \vec{F} = & \frac{1}{r \sin \theta} \left(\frac{\partial(\sin \theta F_\phi)}{\partial \theta} - \frac{\partial F_\theta}{\partial \phi} \right) \hat{r} + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial F_r}{\partial \phi} - \frac{\partial(rF_\phi)}{\partial r} \right) \hat{\theta} \\ & + \frac{1}{r} \left(\frac{\partial(rF_\theta)}{\partial r} - \frac{\partial F_r}{\partial \theta} \right) \hat{\phi} \end{aligned}$$

Example 2.10. Find $\nabla \times \vec{F}$ where $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$.

Solution:

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix} \\ &= (0 - 1)\hat{i} + (1 - 0)\hat{j} + (0 - 1)\hat{k} \\ &= -\hat{i} + \hat{j} - \hat{k} \end{aligned}$$

2.5 The Laplacian

Definition 2.11. The Laplacian of a scalar field ϕ is:

$$\nabla^2 \phi = \nabla \cdot (\nabla \phi) = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

Formula 2.12 (Laplacian in Different Coordinate Systems). **Cartesian:**

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

Cylindrical:

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2}$$

Spherical:

$$\nabla^2 \phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \phi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \phi}{\partial \phi^2}$$

2.6 Important Vector Identities

Important for GATE

Must Memorize for GATE:

$$\nabla \times (\nabla \phi) = 0 \quad (\text{curl of gradient is zero}) \quad (1)$$

$$\nabla \cdot (\nabla \times \vec{F}) = 0 \quad (\text{divergence of curl is zero}) \quad (2)$$

$$\nabla \times (\nabla \times \vec{F}) = \nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F} \quad (3)$$

$$\nabla \cdot (\phi \vec{F}) = \phi(\nabla \cdot \vec{F}) + \vec{F} \cdot (\nabla \phi) \quad (4)$$

$$\nabla \times (\phi \vec{F}) = \phi(\nabla \times \vec{F}) + (\nabla \phi) \times \vec{F} \quad (5)$$

$$\nabla(\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \nabla) \vec{B} + (\vec{B} \cdot \nabla) \vec{A} + \vec{A} \times (\nabla \times \vec{B}) + \vec{B} \times (\nabla \times \vec{A}) \quad (6)$$

$$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B}) \quad (7)$$

$$\nabla \times (\vec{A} \times \vec{B}) = \vec{A}(\nabla \cdot \vec{B}) - \vec{B}(\nabla \cdot \vec{A}) + (\vec{B} \cdot \nabla) \vec{A} - (\vec{A} \cdot \nabla) \vec{B} \quad (8)$$

2.7 Line Integrals

Definition 2.13. The line integral of a vector field $\vec{F}$ along a curve C from point A to point B is:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (F_x dx + F_y dy + F_z dz)$$

Exam Tip

How to Evaluate Line Integrals:

1. Parameterize the curve: $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$, $t \in [a, b]$
2. Find $d\vec{r} = \frac{d\vec{r}}{dt} dt$
3. Express $\vec{F}$ in terms of parameter t
4. Evaluate $\int_a^b \vec{F}(\vec{r}(t)) \cdot \frac{d\vec{r}}{dt} dt$

Example 2.14. Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = y\hat{i} + x\hat{j}$ along the straight line from $(0, 0)$ to $(1, 1)$.

Solution: Parameterize: $\vec{r}(t) = t\hat{i} + t\hat{j}$, $t \in [0, 1]$

$$d\vec{r} = (\hat{i} + \hat{j})dt$$

$$\vec{F}(\vec{r}(t)) = t\hat{i} + t\hat{j}$$

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_0^1 (t\hat{i} + t\hat{j}) \cdot (\hat{i} + \hat{j}) dt \\ &= \int_0^1 2t dt = [t^2]_0^1 = 1 \end{aligned}$$

2.8 Conservative Fields and Potential Functions

Definition 2.15. A vector field $\vec{F}$ is **conservative** if:

- $\vec{F} = \nabla\phi$ for some scalar function ϕ (called potential)
- $\nabla \times \vec{F} = 0$ (irrotational)
- $\oint_C \vec{F} \cdot d\vec{r} = 0$ for any closed path
- Line integral is path-independent

Important for GATE

Test for Conservative Field (in simply-connected region):

$$\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k} \text{ is conservative if } \nabla \times \vec{F} = 0$$

Equivalently:

$$\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x}, \quad \frac{\partial F_y}{\partial z} = \frac{\partial F_z}{\partial y}, \quad \frac{\partial F_z}{\partial x} = \frac{\partial F_x}{\partial z}$$

Example 2.16. Determine if $\vec{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$ is conservative. If yes, find the potential.

Solution: Check: $\frac{\partial F_x}{\partial y} = 2x = \frac{\partial F_y}{\partial x}$

$$\frac{\partial F_y}{\partial z} = 0 = \frac{\partial F_z}{\partial y}$$

$$\frac{\partial F_z}{\partial x} = 3z^2 = \frac{\partial F_x}{\partial z}$$

Conservative! To find ϕ : $\nabla\phi = \vec{F}$

$$\frac{\partial\phi}{\partial x} = 2xy + z^3 \implies \phi = x^2y + xz^3 + f(y, z)$$

$$\frac{\partial\phi}{\partial y} = x^2 + \frac{\partial f}{\partial y} = x^2 \implies f(y, z) = g(z)$$

$$\frac{\partial\phi}{\partial z} = 3xz^2 + g'(z) = 3xz^2 \implies g'(z) = 0$$

Therefore: $\phi = x^2y + xz^3 + C$

2.9 Surface Integrals

Definition 2.17. The surface integral of a vector field $\vec{F}$ over a surface S is:

$$\iint_S \vec{F} \cdot d\vec{A} = \iint_S \vec{F} \cdot \hat{n} dA$$

where $\hat{n}$ is the unit normal to the surface.

Exam Tip

How to Evaluate Surface Integrals:

1. Parameterize the surface: $\vec{r}(u, v)$
2. Find normal vector: $\vec{n} = \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v}$
3. Calculate: $\iint_D \vec{F}(\vec{r}(u, v)) \cdot \vec{n} du dv$

2.10 Fundamental Theorems

Theorem 2.18 (Gradient Theorem). For a scalar field ϕ and curve C from A to B :

$$\int_C \nabla \phi \cdot d\vec{r} = \phi(B) - \phi(A)$$

Theorem 2.19 (Gauss's Divergence Theorem).

$$\iiint_V (\nabla \cdot \vec{F}) dV = \oint_S \vec{F} \cdot d\vec{A}$$

Relates volume integral of divergence to flux through closed surface.

Theorem 2.20 (Stokes' Theorem).

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{A} = \oint_C \vec{F} \cdot d\vec{r}$$

Relates surface integral of curl to line integral around boundary.

Theorem 2.21 (Green's Theorem (2D version of Stokes)).

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \oint_C (P dx + Q dy)$$

Important for GATE

Quick Reference - When to Use Which Theorem:

- **Gradient Theorem:** Line integral of conservative field
- **Divergence Theorem:** Converting volume to surface integral (flux)
- **Stokes' Theorem:** Converting surface to line integral (circulation)
- **Green's Theorem:** 2D problems in plane

2.11 Practice Problems - Vector Calculus

1. Calculate $\nabla \times \vec{F}$ and $\nabla \cdot \vec{F}$ for $\vec{F} = x^2 y \hat{i} + y^2 z \hat{j} + z^2 x \hat{k}$
2. Show that $\vec{F} = (y^2 \cos x + z^3) \hat{i} + (2y \sin x - 4) \hat{j} + (3xz^2 + 2) \hat{k}$ is conservative and find its potential.
3. Verify Stokes' theorem for $\vec{F} = -y \hat{i} + x \hat{j}$ over the hemisphere $x^2 + y^2 + z^2 = 1$, $z \geq 0$.
4. Use divergence theorem to evaluate $\oint_S \vec{F} \cdot d\vec{A}$ where $\vec{F} = x^3 \hat{i} + y^3 \hat{j} + z^3 \hat{k}$ and S is the sphere $x^2 + y^2 + z^2 = a^2$.
5. Find the work done by $\vec{F} = (x^2 + y^2) \hat{i} + 2xy \hat{j}$ along the path $y = x^2$ from $(0, 0)$ to $(1, 1)$.

3 Complex Analysis

3.1 Complex Numbers - Foundation

Definition 3.1. A complex number z is written as:

$$z = x + iy = re^{i\theta}$$

where $x = \operatorname{Re}(z)$, $y = \operatorname{Im}(z)$, $r = |z| = \sqrt{x^2 + y^2}$, $\theta = \arg(z) = \tan^{-1}(y/x)$

Important for GATE

Key Formulas:

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (\text{Euler's formula}) \quad (9)$$

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \quad (10)$$

$$z^* = x - iy = re^{-i\theta} \quad (\text{complex conjugate}) \quad (11)$$

$$|z|^2 = zz^* = x^2 + y^2 \quad (12)$$

$$\frac{1}{z} = \frac{z^*}{|z|^2} \quad (13)$$

3.2 Analytic Functions

Definition 3.2. A function $f(z) = u(x, y) + iv(x, y)$ is **analytic** (holomorphic) at point z_0 if:

- $f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$ exists
- The limit is independent of the path of approach

Theorem 3.3 (Cauchy-Riemann Equations). $f(z) = u(x, y) + iv(x, y)$ is analytic if and only if:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

and all partial derivatives are continuous.

Important for GATE

Testing for Analyticity - GATE Shortcut:

1. Identify $u(x, y)$ and $v(x, y)$
2. Check if C-R equations are satisfied
3. Check continuity of partial derivatives

Example 3.4. Is $f(z) = z^2$ analytic?

Solution: $f(z) = (x + iy)^2 = x^2 - y^2 + 2ixy$

$$u = x^2 - y^2, \quad v = 2xy$$

$$\frac{\partial u}{\partial x} = 2x = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x}$$

Yes, $f(z) = z^2$ is analytic everywhere (entire function).

3.3 Harmonic Functions

Definition 3.5. A function $u(x, y)$ is **harmonic** if it satisfies Laplace's equation:

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Theorem 3.6. If $f(z) = u + iv$ is analytic, then both u and v are harmonic functions, and v is the **harmonic conjugate** of u .

Exam Tip

Finding Harmonic Conjugate: Given harmonic u , to find v :

1. From C-R: $\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x}$, integrate w.r.t. y
2. From C-R: $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$, use to find integration constant

3.4 Complex Integration

Definition 3.7. The contour integral of $f(z)$ along curve C is:

$$\int_C f(z) dz$$

Theorem 3.8 (Cauchy's Integral Theorem). If $f(z)$ is analytic in a simply-connected domain D , then for any closed contour C in D :

$$\oint_C f(z) dz = 0$$

Important for GATE

Implications of Cauchy's Theorem:

- Integral of analytic function around closed loop = 0
- Integral between two points is path-independent
- Can deform contours without changing integral value

Theorem 3.9 (Cauchy's Integral Formula). If $f(z)$ is analytic inside and on a simple closed contour C , and z_0 is inside C :

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$$

For derivatives:

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz$$

Example 3.10. Evaluate $\oint_C \frac{e^z}{z-i} dz$ where C is the circle $|z| = 2$.

Solution: $z_0 = i$ is inside C , and $f(z) = e^z$ is analytic.

By Cauchy's Integral Formula:

$$\oint_C \frac{e^z}{z-i} dz = 2\pi i \cdot e^i = 2\pi i(\cos 1 + i \sin 1)$$

3.5 Taylor and Laurent Series

Theorem 3.11 (Taylor Series). If $f(z)$ is analytic at z_0 , it can be expanded as:

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

where $a_n = \frac{f^{(n)}(z_0)}{n!}$

Theorem 3.12 (Laurent Series). If $f(z)$ is analytic in annulus $r_1 < |z - z_0| < r_2$:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

where:

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz$$

Important for GATE

Types of Singularities:

- **Removable:** Finite number of negative powers (can be removed)
- **Pole of order m :** Laurent series has $(z - z_0)^{-m}$ as highest negative power
- **Essential:** Infinite number of negative powers

3.6 Residue Calculus

Definition 3.13. The **residue** of $f(z)$ at isolated singularity z_0 is the coefficient a_{-1} in Laurent series:

$$\text{Res}(f, z_0) = a_{-1} = \frac{1}{2\pi i} \oint_C f(z) dz$$

Theorem 3.14 (Residue Theorem). If $f(z)$ is analytic inside and on closed contour C except at isolated singularities $z_1, z_2, \dots, z_n$ inside C :

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k)$$

Important for GATE**Formulas for Finding Residues:****1. Simple Pole at z_0 :**

$$\text{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z)$$

2. Pole of Order m at z_0 :

$$\text{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)]$$

3. Function of form $\frac{g(z)}{h(z)}$ with simple zero of h at z_0 :

$$\text{Res}\left(\frac{g}{h}, z_0\right) = \frac{g(z_0)}{h'(z_0)}$$

Example 3.15. Find $\text{Res}\left(\frac{z^2+1}{z(z-1)^2}, 0\right)$ and $\text{Res}\left(\frac{z^2+1}{z(z-1)^2}, 1\right)$.

Solution: At $z = 0$ (simple pole):

$$\text{Res}(f, 0) = \lim_{z \rightarrow 0} z \cdot \frac{z^2 + 1}{z(z-1)^2} = \lim_{z \rightarrow 0} \frac{z^2 + 1}{(z-1)^2} = \frac{1}{1} = 1$$

At $z = 1$ (pole of order 2):

$$\begin{aligned} \text{Res}(f, 1) &= \lim_{z \rightarrow 1} \frac{d}{dz} \left[(z-1)^2 \cdot \frac{z^2 + 1}{z(z-1)^2} \right] = \lim_{z \rightarrow 1} \frac{d}{dz} \left(\frac{z^2 + 1}{z} \right) \\ &= \lim_{z \rightarrow 1} \frac{z \cdot 2z - (z^2 + 1)}{z^2} = \frac{2 - 2}{1} = 0 \end{aligned}$$

3.7 Evaluation of Real Integrals**Exam Tip****Types of Real Integrals Solvable by Residues:**

Type 1: $\int_0^{2\pi} F(\cos \theta, \sin \theta) d\theta$

Substitute: $z = e^{i\theta}$, $\cos \theta = \frac{z+z^{-1}}{2}$, $\sin \theta = \frac{z-z^{-1}}{2i}$, $d\theta = \frac{dz}{iz}$

Contour: $|z| = 1$

Type 2: $\int_{-\infty}^{\infty} f(x) dx$ where $f(z) \rightarrow 0$ as $|z| \rightarrow \infty$ in upper half-plane

Use semicircular contour in upper half-plane, sum residues of poles in upper half-plane.

Type 3: $\int_{-\infty}^{\infty} f(x)e^{iax} dx$ (Fourier integrals)

Use Jordan's lemma with appropriate contour.

Example 3.16. Evaluate $\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta}$.

Solution: Let $z = e^{i\theta}$, then $\cos \theta = \frac{z+z^{-1}}{2}$, $d\theta = \frac{dz}{iz}$

$$\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta} = \oint_{|z|=1} \frac{1}{2 + \frac{z+z^{-1}}{2}} \cdot \frac{dz}{iz} = \oint_{|z|=1} \frac{2 dz}{iz(4 + z + z^{-1})}$$

$$= \oint_{|z|=1} \frac{2 dz}{i(4z + z^2 + 1)} = \oint_{|z|=1} \frac{-2i dz}{z^2 + 4z + 1}$$

Poles: $z = \frac{-4 \pm \sqrt{16-4}}{2} = -2 \pm \sqrt{3}$

Only $z_1 = -2 + \sqrt{3}$ (-0.268) is inside $|z| = 1$.

$$\text{Res}(f, z_1) = \frac{-2i}{2z_1 + 4} = \frac{-2i}{2\sqrt{3}} = \frac{-i}{\sqrt{3}}$$

$$\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta} = 2\pi i \cdot \frac{-i}{\sqrt{3}} = \frac{2\pi}{\sqrt{3}}$$

3.8 Conformal Mapping

Definition 3.17. A mapping $w = f(z)$ is **conformal** at z_0 if it preserves angles between curves and $f'(z_0) \neq 0$.

Important for GATE

Important Conformal Maps:

1. $w = z + c$: Translation
2. $w = az$ (a real): Scaling and rotation
3. $w = \frac{1}{z}$: Inversion
4. $w = z^2$: Squaring
5. $w = e^z$: Exponential
6. $w = \frac{az+b}{cz+d}$: Möbius transformation

3.9 Practice Problems - Complex Analysis

1. Check if $f(z) = |z|^2$ is analytic.
2. Find the harmonic conjugate of $u(x, y) = x^3 - 3xy^2$.
3. Evaluate $\oint_C \frac{z^2+3z}{(z-1)(z-2)} dz$ where $C : |z| = 1.5$.
4. Find all singularities and their types for $f(z) = \frac{e^z-1}{z^3}$.
5. Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$ using residue theorem.
6. Find the Laurent series of $f(z) = \frac{1}{z(z-1)}$ in the annulus $0 < |z| < 1$.
7. Evaluate $\int_0^{2\pi} \frac{d\theta}{5+4\sin \theta}$ using residue calculus.

4 Differential Equations

4.1 Classification of Differential Equations

Definition 4.1. Order: Highest derivative present

Degree: Power of highest derivative (when polynomial in derivatives)

Linear: Can be written as $a_n(x)y^{(n)} + \dots + a_1(x)y' + a_0(x)y = f(x)$

4.2 First-Order ODEs

Important for GATE

Standard Forms:

1. Separable: $\frac{dy}{dx} = f(x)g(y)$

Solution: $\int \frac{dy}{g(y)} = \int f(x) dx$

2. Linear: $\frac{dy}{dx} + P(x)y = Q(x)$

Integrating factor: $\mu(x) = e^{\int P(x)dx}$

Solution: $y = \frac{1}{\mu(x)} \left[\int \mu(x)Q(x)dx + C \right]$

3. Exact: $M(x, y)dx + N(x, y)dy = 0$ where $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Solution: $\int M dx + \int (\text{terms in } N \text{ not involving } x) dy = C$

4. Bernoulli: $\frac{dy}{dx} + P(x)y = Q(x)y^n$

Substitution: $v = y^{1-n}$ converts to linear.

4.3 Second-Order Linear ODEs

Definition 4.2. Standard form: $y'' + p(x)y' + q(x)y = r(x)$

Homogeneous: $r(x) = 0$

Non-homogeneous: $r(x) \neq 0$

Theorem 4.3 (Superposition Principle). If y_1 and y_2 are solutions to homogeneous equation, then:

$$y = c_1y_1 + c_2y_2$$

is also a solution (if y_1, y_2 are linearly independent).

4.3.1 Constant Coefficients

Important for GATE

Equation: $ay'' + by' + cy = 0$

Characteristic equation: $ar^2 + br + c = 0$

Solutions based on roots:

- 1. Real distinct roots r_1, r_2 :** $y = c_1e^{r_1x} + c_2e^{r_2x}$
- 2. Real repeated root r :** $y = (c_1 + c_2x)e^{rx}$
- 3. Complex roots $\alpha \pm i\beta$:** $y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$

4.4 Series Solutions - Power Series Method

Definition 4.4. Assume solution of form:

$$y = \sum_{n=0}^{\infty} a_n x^n$$

Substitute into DE and match coefficients to find recurrence relation for a_n .

Exam Tip

When to Use Power Series:

- Coefficients are analytic functions
- Near ordinary points
- Initial conditions at $x = 0$

4.5 Frobenius Method

Definition 4.5. For equations with regular singular point at $x = 0$, assume:

$$y = x^r \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n x^{n+r}$$

where $a_0 \neq 0$.

Important for GATE

Frobenius Method Steps:

1. Identify the singular point (usually $x = 0$)
2. Substitute $y = \sum a_n x^{n+r}$ into DE
3. Find indicial equation from lowest power of x
4. Solve indicial equation for r
5. Find recurrence relation for a_n
6. Construct solutions based on roots of indicial equation

Theorem 4.6 (Types of Solutions Based on Indicial Roots). Let r_1 and r_2 be roots of indicial equation with $r_1 \geq r_2$:

Case 1: $r_1 - r_2$ **not integer** Two linearly independent solutions: $y_1 = x^{r_1} \sum a_n x^n$, $y_2 = x^{r_2} \sum b_n x^n$

Case 2: $r_1 = r_2$ $y_1 = x^{r_1} \sum a_n x^n$, $y_2 = y_1 \ln x + x^{r_1} \sum c_n x^n$

Case 3: $r_1 - r_2 = \text{positive integer}$ $y_1 = x^{r_1} \sum a_n x^n$, $y_2 = C y_1 \ln x + x^{r_2} \sum b_n x^n$

Example 4.7. Solve $2xy'' + y' + y = 0$ using Frobenius method.

Solution: $x = 0$ is regular singular point.

Assume $y = \sum_{n=0}^{\infty} a_n x^{n+r}$
 $y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$
 $y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$
 Substitute:

$$2x \sum (n+r)(n+r-1) a_n x^{n+r-2} + \sum (n+r) a_n x^{n+r-1} + \sum a_n x^{n+r} = 0$$

$$\sum [2(n+r)(n+r-1) + (n+r)] a_n x^{n+r-1} + \sum a_n x^{n+r} = 0$$

Lowest power (x^{r-1}): $[2r(r-1) + r] a_0 = 0$

Indicial equation: $2r^2 - r = r(2r-1) = 0$

Roots: $r_1 = \frac{1}{2}$, $r_2 = 0$

(Continue to find recurrence relations...)

4.6 Special Functions

4.6.1 Legendre Polynomials

Definition 4.8. Legendre's equation:

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

Solutions are Legendre polynomials $P_n(x)$ when n is non-negative integer.

Important for GATE

Properties of Legendre Polynomials:

1. Rodrigues' formula: $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$
2. $P_n(1) = 1$, $P_n(-1) = (-1)^n$, $P_n(-x) = (-1)^n P_n(x)$
3. Orthogonality: $\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn}$
4. Recurrence: $(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$
5. First few: $P_0 = 1$, $P_1 = x$, $P_2 = \frac{1}{2}(3x^2 - 1)$, $P_3 = \frac{1}{2}(5x^3 - 3x)$

4.6.2 Bessel Functions

Definition 4.9. Bessel's equation:

$$x^2 y'' + xy' + (x^2 - n^2)y = 0$$

Solutions are Bessel functions $J_n(x)$ (first kind) and $Y_n(x)$ (second kind).

Important for GATE**Properties of Bessel Functions:**

1. Series: $J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(n+k)!} \left(\frac{x}{2}\right)^{2k+n}$
2. $J_{-n}(x) = (-1)^n J_n(x)$ for integer n
3. Orthogonality: $\int_0^a x J_n(\alpha_m x) J_n(\alpha_n x) dx = \frac{a^2}{2} [J_{n+1}(\alpha_m a)]^2 \delta_{mn}$
4. Recurrence: $J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x)$
5. Derivative: $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$

4.6.3 Hermite Polynomials

Definition 4.10. Hermite's equation:

$$y'' - 2xy' + 2ny = 0$$

Solutions are Hermite polynomials $H_n(x)$ when n is non-negative integer.

Formula 4.11 (Hermite Polynomial Properties). 1. Generating function: $e^{2xt-t^2} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n$

2. Rodrigues: $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$
3. Orthogonality: $\int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx = \sqrt{\pi} 2^n n! \delta_{mn}$
4. First few: $H_0 = 1$, $H_1 = 2x$, $H_2 = 4x^2 - 2$, $H_3 = 8x^3 - 12x$

4.6.4 Laguerre Polynomials

Definition 4.12. Laguerre's equation:

$$xy'' + (1-x)y' + ny = 0$$

Solutions are Laguerre polynomials $L_n(x)$.

Formula 4.13 (Laguerre Polynomial Properties). 1. Rodrigues: $L_n(x) = \frac{e^x}{n!} \frac{d^n}{dx^n} (x^n e^{-x})$

2. Orthogonality: $\int_0^{\infty} L_m(x) L_n(x) e^{-x} dx = \delta_{mn}$
3. Associated Laguerre: $L_n^k(x) = (-1)^k \frac{d^k}{dx^k} L_{n+k}(x)$ (appears in hydrogen atom)

4.7 Sturm-Liouville Theory

Definition 4.14. Sturm-Liouville equation:

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda w(x)] y = 0$$

with boundary conditions.

- Theorem 4.15** (Properties of Sturm-Liouville Problems). 1. Eigenvalues are real
2. Eigenfunctions corresponding to different eigenvalues are orthogonal with weight $w(x)$
3. Eigenfunctions form a complete set

4.8 Partial Differential Equations - Separation of Variables

Exam Tip

Method of Separation of Variables:

1. Assume $u(x, t) = X(x)T(t)$ (or appropriate variables)
2. Substitute into PDE
3. Separate variables: each side equals constant
4. Solve resulting ODEs with boundary conditions
5. Superpose solutions to satisfy initial conditions

4.8.1 Heat Equation

Example 4.16. Solve: $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$, $0 < x < L$

BC: $u(0, t) = u(L, t) = 0$, IC: $u(x, 0) = f(x)$

Solution: Assume $u(x, t) = X(x)T(t)$:

$$X(x)T'(t) = kX''(x)T(t) \implies \frac{T'}{kT} = \frac{X''}{X} = -\lambda$$

$X'' + \lambda X = 0$ with $X(0) = X(L) = 0$ gives:

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, X_n = \sin \frac{n\pi x}{L}$$

$$T' + k\lambda_n T = 0 \text{ gives } T_n = e^{-k\lambda_n t}$$

General solution:

$$u(x, t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} e^{-k(n\pi/L)^2 t}$$

where $b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$

4.8.2 Wave Equation

Formula 4.17. $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$

Solution: $u(x, t) = \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi ct}{L} + b_n \sin \frac{n\pi ct}{L} \right) \sin \frac{n\pi x}{L}$

4.8.3 Laplace Equation

Formula 4.18. $\nabla^2 u = 0$ (in 2D: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$)

Solutions in different geometries use separation in appropriate coordinates.

4.9 Green's Functions

Definition 4.19. Green's function $G(x, \xi)$ for operator L satisfies:

$$LG(x, \xi) = \delta(x - \xi)$$

with appropriate boundary conditions.

Solution to $Ly = f(x)$ is: $y(x) = \int G(x, \xi)f(\xi)d\xi$

4.10 Practice Problems - Differential Equations

1. Solve $x\frac{dy}{dx} + 2y = x^2$ using integrating factor.
2. Find general solution: $y'' - 4y' + 4y = 0$
3. Solve by Frobenius method: $xy'' + 2y' - y = 0$
4. Express $f(x) = x$ on $[-1, 1]$ as series of Legendre polynomials.
5. Show that $J_0(x)$ satisfies $xJ_0'' + J_0' + xJ_0 = 0$.
6. Solve heat equation with $u(x, 0) = \sin(3\pi x/L)$ on $[0, L]$.

5 Linear Algebra

5.1 Vector Spaces

Definition 5.1. A **vector space** V over field $\mathbb{F}$ is a set with operations of addition and scalar multiplication satisfying:

1. Closure under addition and scalar multiplication
2. Associativity and commutativity of addition
3. Existence of zero vector and additive inverses
4. Distributivity and compatibility of scalar multiplication

Definition 5.2. Subspace: Subset $W \subseteq V$ that is itself a vector space.

Span: $\text{span}\{v_1, \dots, v_n\} = \{c_1v_1 + \dots + c_nv_n : c_i \in \mathbb{F}\}$

Linear Independence: Vectors $v_1, \dots, v_n$ are linearly independent if:

$$c_1v_1 + \dots + c_nv_n = 0 \implies c_1 = \dots = c_n = 0$$

Basis: Linearly independent spanning set.

Dimension: Number of vectors in basis.

5.2 Matrices and Determinants

Important for GATE

Matrix Operations:

1. Addition: $(A + B)_{ij} = A_{ij} + B_{ij}$
2. Scalar multiplication: $(cA)_{ij} = cA_{ij}$
3. Multiplication: $(AB)_{ij} = \sum_k A_{ik}B_{kj}$
4. Transpose: $(A^T)_{ij} = A_{ji}$
5. Conjugate transpose (Hermitian): $(A^\dagger)_{ij} = \overline{A_{ji}}$

Formula 5.3 (Properties of Determinants). 1. $\det(AB) = \det(A)\det(B)$

2. $\det(A^T) = \det(A)$

3. $\det(A^{-1}) = \frac{1}{\det(A)}$

4. $\det(cA) = c^n \det(A)$ for $n \times n$ matrix

5. Row operations: swap rows $\rightarrow$ sign change; add multiple of row $\rightarrow$ no change; multiply row by $c \rightarrow$ multiply det by c

Exam Tip**Computing Determinants:**

- 2×2 : $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$
- 3×3 : Use cofactor expansion or Sarrus rule
- General: Use row reduction to triangular form, then product of diagonal

5.3 Eigenvalues and Eigenvectors

Definition 5.4. For matrix A , scalar λ is an **eigenvalue** with **eigenvector** $v \neq 0$ if:

$$Av = \lambda v$$

Important for GATE**Finding Eigenvalues and Eigenvectors:**

1. Find characteristic polynomial: $\det(A - \lambda I) = 0$
2. Solve for eigenvalues $\lambda_1, \dots, \lambda_n$
3. For each λ_i , solve $(A - \lambda_i I)v = 0$ for eigenvector v

Theorem 5.5 (Properties of Eigenvalues). For $n \times n$ matrix A with eigenvalues $\lambda_1, \dots, \lambda_n$:

1. $\text{tr}(A) = \sum \lambda_i$
2. $\det(A) = \prod \lambda_i$
3. A and A^T have same eigenvalues
4. Eigenvalues of A^k are λ_i^k
5. Eigenvalues of A^{-1} are λ_i^{-1} (if A invertible)

Example 5.6. Find eigenvalues and eigenvectors of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

Solution: Characteristic equation:

$$\det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = 0$$

Eigenvalues: $\lambda_1 = 3, \lambda_2 = 1$

For $\lambda_1 = 3$:

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \implies v^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $\lambda_2 = 1$:

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \implies v^{(2)} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

5.4 Diagonalization

Definition 5.7. Matrix A is **diagonalizable** if there exists invertible P such that:

$$P^{-1}AP = D$$

where D is diagonal.

Theorem 5.8. A is diagonalizable if and only if A has n linearly independent eigenvectors.

If $P = [v_1 | v_2 | \cdots | v_n]$ (eigenvectors as columns), then:

$$P^{-1}AP = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$$

Important for GATE

Applications of Diagonalization:

1. Computing matrix powers: $A^n = PD^nP^{-1}$
2. Solving systems of ODEs
3. Finding matrix exponential: $e^A = Pe^DP^{-1}$
4. Understanding dynamics of linear systems

5.5 Hermitian and Unitary Matrices

Definition 5.9. Hermitian matrix: $A^\dagger = A$ (self-adjoint)

Unitary matrix: $U^\dagger U = UU^\dagger = I$

Real versions:

- Symmetric: $A^T = A$
- Orthogonal: $O^T O = O O^T = I$

Theorem 5.10 (Spectral Theorem for Hermitian Matrices). If A is Hermitian, then:

1. All eigenvalues are real
2. Eigenvectors corresponding to distinct eigenvalues are orthogonal
3. A can be diagonalized by unitary matrix: $U^\dagger A U = D$
4. $A = \sum_i \lambda_i |v_i\rangle\langle v_i|$ (spectral decomposition)

Important for GATE

Why Hermitian Matrices Matter in Physics:

- Observables in quantum mechanics are Hermitian operators
- Real eigenvalues = measurable quantities
- Orthogonal eigenvectors = orthogonal states
- Complete set of eigenvectors = complete set of states

Example 5.11. Show that $A = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$ is Hermitian and find its eigenvalues.

Solution: Check: $A^\dagger = \begin{pmatrix} 1 & -(-i) \\ -i & 1 \end{pmatrix} = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} = A$

Characteristic equation:

$$\det \begin{pmatrix} 1-\lambda & i \\ -i & 1-\lambda \end{pmatrix} = (1-\lambda)^2 + 1 = \lambda^2 - 2\lambda + 2 = 0$$

Wait, this gives complex eigenvalues! Let me recalculate...

Actually: $(1-\lambda)^2 - (i)(-i) = (1-\lambda)^2 - 1 = -2\lambda + \lambda^2 = 0$

$\lambda(\lambda-2) = 0$, so $\lambda = 0, 2$ (real, as expected)

5.6 Inner Product Spaces

Definition 5.12. An **inner product** on vector space V is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$ satisfying:

1. Conjugate symmetry: $\langle u, v \rangle = \overline{\langle v, u \rangle}$
2. Linearity in first argument: $\langle au + bv, w \rangle = a\langle u, w \rangle + b\langle v, w \rangle$
3. Positive definiteness: $\langle v, v \rangle \geq 0$ with equality iff $v = 0$

Formula 5.13 (Important Inner Products). **Euclidean:** $\langle x, y \rangle = x^T y = \sum x_i y_i$

Complex: $\langle x, y \rangle = x^\dagger y = \sum \bar{x}_i y_i$

Function space L^2 : $\langle f, g \rangle = \int f^*(x)g(x)dx$

Definition 5.14. Norm: $\|v\| = \sqrt{\langle v, v \rangle}$

Orthogonal: $\langle u, v \rangle = 0$

Orthonormal basis: $\langle e_i, e_j \rangle = \delta_{ij}$

5.7 Gram-Schmidt Orthogonalization

Important for GATE

Gram-Schmidt Process:

Given linearly independent vectors $v_1, \dots, v_n$, construct orthonormal $e_1, \dots, e_n$:

$$u_1 = v_1 \tag{14}$$

$$u_2 = v_2 - \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 \tag{15}$$

$$u_3 = v_3 - \frac{\langle v_3, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 - \frac{\langle v_3, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 \tag{16}$$

$$\vdots \tag{17}$$

$$e_i = \frac{u_i}{\|u_i\|} \tag{18}$$

5.8 Linear Transformations

Definition 5.15. A function $T : V \rightarrow W$ is a **linear transformation** if:

$$T(au + bv) = aT(u) + bT(v)$$

Definition 5.16. Kernel (null space): $\ker(T) = \{v \in V : T(v) = 0\}$

Image (range): $\text{Im}(T) = \{T(v) : v \in V\}$

Rank: $\text{rank}(T) = \dim(\text{Im}(T))$

Nullity: $\text{nullity}(T) = \dim(\ker(T))$

Theorem 5.17 (Rank-Nullity Theorem). For linear transformation $T : V \rightarrow W$:

$$\dim(V) = \text{rank}(T) + \text{nullity}(T)$$

5.9 Change of Basis

Formula 5.18. If $[v]_B$ represents vector v in basis B and $[v]_{B'}$ in basis B' :

$$[v]_{B'} = P_{B \rightarrow B'}[v]_B$$

where $P_{B \rightarrow B'}$ is change of basis matrix.

For linear transformation T with matrix $[T]_B$ in basis B :

$$[T]_{B'} = P^{-1}[T]_B P$$

5.10 Special Matrices and Decompositions

Important for GATE

Important Matrix Decompositions:

1. **LU Decomposition:** $A = LU$ (lower $\times$ upper triangular)
2. **QR Decomposition:** $A = QR$ (orthogonal $\times$ upper triangular)
3. **Singular Value Decomposition (SVD):** $A = U\Sigma V^\dagger$
 - U, V unitary
 - Σ diagonal with singular values $\sigma_i \geq 0$
 - Works for any matrix (not just square)
4. **Spectral Decomposition (Hermitian):** $A = \sum_i \lambda_i |v_i\rangle\langle v_i|$

5.11 Cayley-Hamilton Theorem

Theorem 5.19. Every square matrix satisfies its own characteristic equation.

If $p(\lambda) = \det(A - \lambda I)$ is the characteristic polynomial, then $p(A) = 0$.

Exam Tip**Use of Cayley-Hamilton:**

- Computing matrix powers
- Finding matrix inverse
- Simplifying matrix polynomials

5.12 Positive Definite Matrices

Definition 5.20. Real symmetric (or Hermitian) matrix A is **positive definite** if:

$$x^T A x > 0 \quad \forall x \neq 0$$

Theorem 5.21 (Equivalent Conditions). For Hermitian A , the following are equivalent:

1. A is positive definite
2. All eigenvalues are positive
3. All principal minors are positive
4. $A = B^\dagger B$ for some invertible B

5.13 Practice Problems - Linear Algebra

1. Find eigenvalues and eigenvectors of $A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$. Is it diagonalizable?
2. Show that eigenvalues of unitary matrix have magnitude 1.
3. Prove that trace of product is independent of order: $\text{tr}(AB) = \text{tr}(BA)$.
4. Use Gram-Schmidt to orthonormalize: $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$, $v_3 = (0, 1, 1)$.
5. Find SVD of $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.
6. Verify Cayley-Hamilton theorem for $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.
7. Show that if A is Hermitian and B is unitary, then $BAB^\dagger$ is Hermitian.

6 GATE Exam Strategy

6.1 Time Management

Exam Tip

3-Hour Exam Strategy:

- **First 15 min:** Quick scan of all questions
- **Next 90 min:** Solve questions you're confident about
- **Next 60 min:** Tackle moderate difficulty questions
- **Last 15 min:** Quick check and difficult questions

6.2 Common Mistakes to Avoid

1. Not checking dimensions in vector calculus
2. Sign errors in determinants and cross products
3. Forgetting $2\pi i$ in Cauchy's formulas
4. Wrong coordinate system formulas (cylindrical vs spherical)
5. Algebraic errors in eigenvalue calculations
6. Not verifying orthogonality conditions
7. Boundary condition mistakes in PDEs

6.3 Formula Sheet - Quick Reference

Important for GATE

Vector Calculus:

$$\begin{aligned}\nabla\phi &= \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k} \\ \nabla \cdot \vec{F} &= \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \\ \nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ F_x & F_y & F_z \end{vmatrix}\end{aligned}$$

Complex Analysis:

$$\begin{aligned}f(z_0) &= \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz \\ \oint_C f(z) dz &= 2\pi i \sum \text{Res}(f, z_k)\end{aligned}$$

Special Functions:

$$\begin{aligned}P_n(1) &= 1, \quad P_n(-1) = (-1)^n \\ \int_{-1}^1 P_m P_n dx &= \frac{2}{2n+1} \delta_{mn}\end{aligned}$$

Linear Algebra:

$$\begin{aligned}\det(AB) &= \det(A) \det(B) \\ \text{tr}(A) &= \sum \lambda_i \\ A = PDP^{-1} &\implies A^n = PD^nP^{-1}\end{aligned}$$

6.4 Previous Years' Focus Areas

1. Divergence theorem applications
2. Residue calculus for real integrals
3. Frobenius method for ODEs
4. Legendre and Bessel function properties
5. Eigenvalue problems
6. Hermitian matrix properties
7. Orthogonalization procedures

6.5 Final Tips

Exam Tip

Last Month Before Exam:

- Solve 10+ previous years' papers under timed conditions
- Revise this formula sheet daily
- Focus on speed and accuracy
- Practice standard problems (each topic 20+ problems)
- Don't learn new topics - consolidate what you know
- Stay healthy - sleep well, eat well, exercise

7 Conclusion

Mathematical Physics is the backbone of GATE Physics. Master these four pillars:

1. **Vector Calculus:** div, grad, curl, and integral theorems
2. **Complex Analysis:** Cauchy's theorems and residue calculus
3. **Differential Equations:** Frobenius method and special functions
4. **Linear Algebra:** Eigenvalues and Hermitian matrices

Important for GATE

Your Success Mantra:

Practice + Understanding + Speed = GATE Success

Study this guide thoroughly. Solve all examples. Practice problems daily. You will ace GATE 2026!

All the Best for GATE 2026!